```
const { Document, Paragraph, TextRun, HeadingLevel, AlignmentType, Table, TableRow, TableCell,
WidthType, BorderStyle, PageBreak, NumberFormat } = require('docx');
const fs = require('fs');
const { Packer } = require('docx');

// ARK SI PROMPTWARE DESIGN DOCUMENT - INTEGRATED VISUALIZATION EDITION

const doc = new Document({
  sections: [{
    properties: {},
    children: [
      // COVER PAGE
      new Paragraph({
        alignment: AlignmentType.CENTER,
        spacing: { before: 2000, after: 400 },
        children: [
          new TextRun({
            text: "ARK SI PROMPTWARE DESIGN DOCUMENT",
            bold: true,
            size: 36,
            font: "Calibri"
          })
        ]
      }),
      new Paragraph({
        alignment: AlignmentType.CENTER,
        spacing: { after: 200 },
        children: [
          new TextRun({
            text: "Integrated Visualization Edition v2.0",
            size: 28,
            font: "Calibri",
            color: "2E75B6"
          })
        ]
      }),
      new Paragraph({
        alignment: AlignmentType.CENTER,
        spacing: { after: 800 },
        children: [
          new TextRun({
            text: "Career Intelligence & AI Vulnerability Platform",
            size: 20,
            italics: true,
            font: "Calibri"
          })
        ]
      }),
```

```
// META INFORMATION TABLE
new Table({
  width: { size: 100, type: WidthType.PERCENTAGE },
  rows: [
    new TableRow({
      children: [
        new TableCell({
          width: { size: 30, type: WidthType.PERCENTAGE },
          children: [new Paragraph({ text: "Document Type:", bold: true })],
        }),
        new TableCell({
          width: { size: 70, type: WidthType.PERCENTAGE },
          children: [new Paragraph("ATLAS PromptWare Design Document (PDD)")],
        }),
      ],
    }),
    new TableRow({
      children: [
        new TableCell({
          children: [new Paragraph({ text: "Version:", bold: true })],
        }),
        new TableCell({
          children: [new Paragraph("2.0 - Integrated Visualization Edition")],
        }),
      ],
    }),
    new TableRow({
      children: [
        new TableCell({
          children: [new Paragraph({ text: "Date:", bold: true })],
        }),
        new TableCell({
          children: [new Paragraph("January 3, 2026")],
        }),
      ],
    }),
    new TableRow({
      children: [
        new TableCell({
          children: [new Paragraph({ text: "Classification:", bold: true })],
        }),
        new TableCell({
          children: [new Paragraph("Enterprise-Grade SPC/PDD GPT")],
        }),
      ],
    }),
    new TableRow({
      children: [
        new TableCell({
```

```
              children: [new Paragraph({ text: "JCSE Score:", bold: true })],
            }),
            new TableCell({
              children: [new Paragraph({ text: "48/50 (FORGE Ultra Premium)", color: "00B050"
})],
            }),
          ],
        }),
        new TableRow({
          children: [
            new TableCell({
              children: [new Paragraph({ text: "Primary SPC:", bold: true })],
            }),
            new TableCell({
              children: [new Paragraph("HR Specialist + AI/ML Expert + Data Scientist + Career
Coach + Market Analyst")],
            }),
          ],
        }),
        new TableRow({
          children: [
            new TableCell({
              children: [new Paragraph({ text: "Visualization Platforms:", bold: true })],
            }),
            new TableCell({
              children: [new Paragraph("NotebookLM + Canva + Native PDF Generation +
Interactive Dashboards")],
            }),
          ],
        }),
        new TableRow({
          children: [
            new TableCell({
              children: [new Paragraph({ text: "Token Optimization:", bold: true })],
            }),
            new TableCell({
              children: [new Paragraph("45% reduction via ZPOS Level 3 | 97% semantic
preservation")],
            }),
          ],
        }),
      ],
    }),

    new PageBreak(),

    // ============================================================
    // EXECUTIVE SUMMARY - PLAIN ENGLISH ANALYTICS
    // ============================================================
    new Paragraph({
```

```
        heading: HeadingLevel.HEADING_1,
        spacing: { before: 400, after: 200 },
        children: [
          new TextRun({
            text: "EXECUTIVE SUMMARY",
            bold: true,
            size: 32,
            color: "2E75B6"
          })
        ]
      }),

      new Paragraph({
        spacing: { before: 200, after: 200 },
        alignment: AlignmentType.JUSTIFIED,
        children: [
          new TextRun({
            text: "The ARK SI (Strategic Intelligence) platform represents a breakthrough in
career intelligence and AI vulnerability assessment, transforming how professionals navigate the
rapidly evolving AI-augmented economy. Deployed across a Fortune 500 consulting firm with 12,000
employees, ARK SI achieved remarkable results: reducing attrition from 18% to 11% (39%
improvement), generating $6M in annual cost savings, and delivering an 8,955% ROI within the
first year. The platform analyzes career value through the JST Index framework (Jobs-Skills-
Talent layers), assesses AI displacement risk using a scientifically validated 5-level
vulnerability meter (0-4 scale), and maps skill transferability vectors to identify optimal
career pivots. With 8,742 completed assessments averaging just 12 minutes each (compared to 4-6
hours for traditional career coaching), ARK SI maintains a 4.6/5.0 satisfaction score and 64%
quarterly repeat usage. The integrated visualization engine now leverages NotebookLM's Nano
Banana Pro model, Canva's design API, and custom PDF generation to produce professional
infographics, interactive dashboards, and comprehensive career intelligence reports. Built on
the ATLAS PromptWare methodology with 98 optimized prompt modules achieving 45% token reduction
while maintaining 97% semantic accuracy, ARK SI exemplifies how systematic prompt engineering
transforms conventional GPT applications into enterprise-grade operational intelligence systems.
The platform's success demonstrates that AI vulnerability is not about job elimination but
strategic career positioning, enabling professionals to quantify their market value, understand
automation risks, and execute data-driven upskilling strategies that ensure long-term career
resilience in an AI-native economy.",
            size: 24,
            font: "Calibri"
          })
        ]
      }),

      new PageBreak(),

      // ============================================================
      // ARK SI SINGLE-PAGE VISUAL SUMMARY SPECIFICATIONS
      // ============================================================
      new Paragraph({
        heading: HeadingLevel.HEADING_1,
```

```
    spacing: { before: 400, after: 200 },
    children: [
      new TextRun({
        text: "ARK SI SINGLE-PAGE VISUAL SUMMARY SPECIFICATIONS",
        bold: true,
        size: 28,
        color: "2E75B6"
      })
    ]
  }),

  new Paragraph({
    heading: HeadingLevel.HEADING_2,
    spacing: { before: 300, after: 150 },
    children: [
      new TextRun({
        text: "1. Visual Summary Architecture",
        bold: true,
        size: 24
      })
    ]
  }),

  new Paragraph({
    spacing: { before: 100, after: 100 },
    children: [
      new TextRun({
        text: "Format Specifications:",
        bold: true,
        size: 22
      })
    ]
  }),

  new Paragraph({
    spacing: { before: 50, after: 50, left: 300 },
    bullet: { level: 0 },
    children: [
      new TextRun("Output Formats: PDF (primary), PNG, SVG, Interactive HTML Dashboard")
    ]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 300 },
    bullet: { level: 0 },
    children: [
      new TextRun("Page Size: Letter (8.5\" × 11\") in Portrait orientation")
    ]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 300 },
```

```
          bullet: { level: 0 },
          children: [
            new TextRun("Color Scheme: Junglenomics brand (Primary: #2E75B6, Secondary: #00B050,
Accent: #FFC000)")
          ]
        }),
        new Paragraph({
          spacing: { before: 50, after: 50, left: 300 },
          bullet: { level: 0 },
          children: [
            new TextRun("Typography: Calibri (headers), Arial (body), Consolas (data)")
          ]
        }),
        new Paragraph({
          spacing: { before: 50, after: 50, left: 300 },
          bullet: { level: 0 },
          children: [
            new TextRun("Layout: 6-section grid system (see layout specifications below)")
          ]
        }),

        new Paragraph({
          heading: HeadingLevel.HEADING_3,
          spacing: { before: 200, after: 100 },
          children: [
            new TextRun({
              text: "1.1 Six-Section Layout Grid",
              bold: true,
              size: 22
            })
          ]
        }),

        // SECTION 1
        new Paragraph({
          spacing: { before: 100, after: 50, left: 300 },
          children: [
            new TextRun({
              text: "SECTION 1: Professional Identity Header (Top 15%)",
              bold: true,
              underline: {}
            })
          ]
        }),
        new Paragraph({
          spacing: { before: 50, after: 50, left: 600 },
          bullet: { level: 1 },
          children: [new TextRun("Name, Title, Industry, Years of Experience")]
        }),
        new Paragraph({
```

```
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Assessment Date & Report ID")]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("QR Code linking to full interactive report")]
  }),

  // SECTION 2
  new Paragraph({
    spacing: { before: 100, after: 50, left: 300 },
    children: [
      new TextRun({
        text: "SECTION 2: JST Index Dashboard (Left Column, 35% width, 40% height)",
        bold: true,
        underline: {}
      })
    ]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Jobs Layer Score (0-100) with market demand indicator")]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Skills Layer Score (0-100) with AI-literacy bonus")]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Talent Layer Score (0-100) with achievement quality")]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Overall JST Index with percentile ranking")]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Radar chart visualization of three layers")]
  }),

  // SECTION 3
  new Paragraph({
    spacing: { before: 100, after: 50, left: 300 },
```

```
  children: [
    new TextRun({
      text: "SECTION 3: AI Vulnerability Meter (Right Column, 35% width, 40% height)",
      bold: true,
      underline: {}
    })
  ]
}),
new Paragraph({
  spacing: { before: 50, after: 50, left: 600 },
  bullet: { level: 1 },
  children: [new TextRun("5-Level color-coded scale (0-4)")]
}),
new Paragraph({
  spacing: { before: 50, after: 50, left: 600 },
  bullet: { level: 1 },
  children: [new TextRun("Level 0 (Red): Critical Displacement (6-18 months)")]
}),
new Paragraph({
  spacing: { before: 50, after: 50, left: 600 },
  bullet: { level: 1 },
  children: [new TextRun("Level 1 (Orange): High Vulnerability (18-36 months)")]
}),
new Paragraph({
  spacing: { before: 50, after: 50, left: 600 },
  bullet: { level: 1 },
  children: [new TextRun("Level 2 (Yellow): Emerging Vulnerability (3-7 years)")]
}),
new Paragraph({
  spacing: { before: 50, after: 50, left: 600 },
  bullet: { level: 1 },
  children: [new TextRun("Level 3 (Blue): Resistant (7-15 years)")]
}),
new Paragraph({
  spacing: { before: 50, after: 50, left: 600 },
  bullet: { level: 1 },
  children: [new TextRun("Level 4 (Green): Resilient & AI-Enhanced (15+ years)")]
}),
new Paragraph({
  spacing: { before: 50, after: 50, left: 600 },
  bullet: { level: 1 },
  children: [new TextRun("Heat map of task-level automation risk")]
}),

// SECTION 4
new Paragraph({
  spacing: { before: 100, after: 50, left: 300 },
  children: [
    new TextRun({
      text: "SECTION 4: Transferability Vector Network (Center, 30% width, 25% height)",
```

```
        bold: true,
        underline: {}
      })
    ]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Network diagram showing current role (center node)")]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Adjacent roles (satellite nodes) with skill gap percentages")]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Edge thickness = transferability score")]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Node size = compensation premium potential")]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Top 5 pivot opportunities highlighted")]
  }),

  // SECTION 5
  new Paragraph({
    spacing: { before: 100, after: 50, left: 300 },
    children: [
      new TextRun({
        text: "SECTION 5: Skills Ecosystem & Recommendations (Bottom 30%)",
        bold: true,
        underline: {}
      })
    ]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Skill inventory matrix (Current vs. Desired)")
    ]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
```

```
    bullet: { level: 1 },
    children: [new TextRun("Context Craft certification pathway visualization")]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Priority upskilling recommendations (New/Up/Ready Skilling)")]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Timeline roadmap with milestones and checkpoints")]
  }),

  // SECTION 6
  new Paragraph({
    spacing: { before: 100, after: 50, left: 300 },
    children: [
      new TextRun({
        text: "SECTION 6: Key Analytics Summary (Footer bar, 100% width × 10% height)",
        bold: true,
        underline: {}
      })
    ]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Market Value Range ($XXK - $YYK)")
    ]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Career Resilience Score (0-100)")]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Estimated Timeline to Next Career Milestone")]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("Recommended Action: [Primary Next Step]")]
  }),

  new PageBreak(),

  // ============================================================
```

```
    // NOTEBOOKLM INTEGRATION ARCHITECTURE
    // ============================================================
    new Paragraph({
      heading: HeadingLevel.HEADING_1,
      spacing: { before: 400, after: 200 },
      children: [
        new TextRun({
          text: "NOTEBOOKLM INTEGRATION ARCHITECTURE",
          bold: true,
          size: 28,
          color: "2E75B6"
        })
      ]
    }),

    new Paragraph({
      heading: HeadingLevel.HEADING_2,
      spacing: { before: 300, after: 150 },
      children: [
        new TextRun({
          text: "2. Nano Banana Pro Integration",
          bold: true,
          size: 24
        })
      ]
    }),

    new Paragraph({
      spacing: { before: 100, after: 100 },
      alignment: AlignmentType.JUSTIFIED,
      children: [
        new TextRun({
          text: "ARK SI leverages Google's Nano Banana Pro model (integrated into NotebookLM
as of November 2025) for automated infographic generation. This integration enables conversion
of dense career intelligence data into professional visual artifacts without manual design
work.",
          size: 22
        })
      ]
    }),

    new Paragraph({
      heading: HeadingLevel.HEADING_3,
      spacing: { before: 200, after: 100 },
      children: [
        new TextRun({
          text: "2.1 NotebookLM Workflow Integration",
          bold: true,
          size: 22
        })
```

```
    ]
  }),

  new Paragraph({
    spacing: { before: 50, after: 50, left: 300 },
    bullet: { level: 0 },
    children: [
      new TextRun({
        text: "Step 1: Data Export to NotebookLM Format",
        bold: true
      })
    ]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 600 },
    bullet: { level: 1 },
    children: [new TextRun("ARK SI generates structured markdown files containing:")
    ]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 900 },
    bullet: { level: 2 },
    children: [new TextRun("JST Index scores with layer breakdowns")]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 900 },
    bullet: { level: 2 },
    children: [new TextRun("AI Vulnerability assessment with timeline projections")]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 900 },
    bullet: { level: 2 },
    children: [new TextRun("Transferability vectors with skill gap analysis")]
  }),
  new Paragraph({
    spacing: { before: 50, after: 50, left: 900 },
    bullet: { level: 2 },
    children: [new TextRun("Personalized recommendations with certification pathways")]
  }),

  new Paragraph({
    spacing: { before: 100, after: 50, left: 300 },
    bullet: { level: 0 },
    children: [
      new TextRun({
        text: "Step 2: Custom Prompt Engineering for Visual Generation",
        bold: true
      })
    ]
  }),
```

```
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Infographic Orientation: Portrait (9:16) for comprehensive
single-page summaries")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Detail Level: Detailed (beta) for maximum information density")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Custom Prompt Template:")
        ]
      }),

      new Paragraph({
        spacing: { before: 100, after: 100, left: 900 },
        shading: { fill: "F2F2F2" },
        border: {
          top: { style: BorderStyle.SINGLE, size: 1, color: "999999" },
          bottom: { style: BorderStyle.SINGLE, size: 1, color: "999999" },
          left: { style: BorderStyle.SINGLE, size: 1, color: "999999" },
          right: { style: BorderStyle.SINGLE, size: 1, color: "999999" }
        },
        children: [
          new TextRun({
            text: "\"Create a professional career intelligence infographic for [CANDIDATE NAME]
using these specifications:\n\nDESIGN STYLE:\n- Junglenomics brand colors (#2E75B6 primary,
#00B050 secondary, #FFC000 accent)\n- Clean, data-driven layout with clear visual hierarchy\n-
Professional typography suitable for C-suite presentation\n- Infographic style with hard-
hitting, quantified insights\n\nCONTENT STRUCTURE:\n1. Header: Professional identity with JST
Index overall score\n2. Left Column: JST three-layer radar chart with numeric scores\n3. Right
Column: AI Vulnerability meter with color-coded 5-level scale\n4. Center: Transferability
network diagram showing career pivot opportunities\n5. Bottom Section: Skills ecosystem matrix
with Context Craft pathway\n6. Footer: Key analytics summary bar\n\nVISUAL ELEMENTS:\n- Use
gauge/meter visualizations for vulnerability assessment\n- Network graph for transferability
vectors\n- Heat maps for skill gap analysis\n- Timeline roadmap for recommended actions\n- All
statistics must match source data exactly\n\nTARGET AUDIENCE: HR executives, career coaches, and
professionals seeking data-driven career guidance\"",
            font: "Consolas",
            size: 18
          })
        ]
      }),

      new Paragraph({
        spacing: { before: 100, after: 50, left: 300 },
```

```
        bullet: { level: 0 },
        children: [
          new TextRun({
            text: "Step 3: Multi-Format Export Pipeline",
            bold: true
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PNG Download: High-resolution (300 DPI) for print quality")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Share Link: Public URL for social media distribution (LinkedIn,
Twitter)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PDF Embedding: Automated integration into comprehensive ARK SI
reports")]
      }),

      new Paragraph({
        heading: HeadingLevel.HEADING_3,
        spacing: { before: 200, after: 100 },
        children: [
          new TextRun({
            text: "2.2 Complementary Visualization Platforms",
            bold: true,
            size: 22
          })
        ]
      }),

      new Paragraph({
        spacing: { before: 100, after: 50, left: 300 },
        children: [
          new TextRun({
            text: "CANVA API INTEGRATION",
            bold: true,
            underline: {}
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
```

```
        bullet: { level: 1 },
        children: [new TextRun("Use Case: Branded templates for enterprise clients requiring
custom styling")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Integration: RESTful API for programmatic design generation")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Advantages: Brand kit compliance, team collaboration features,
extensive template library")]
      }),

      new Paragraph({
        spacing: { before: 100, after: 50, left: 300 },
        children: [
          new TextRun({
            text: "NATIVE PDF GENERATION (Python ReportLab)",
            bold: true,
            underline: {}
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Use Case: Programmatic report generation with precise layout
control")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Technology Stack: ReportLab + Matplotlib for charts + Pandas for
data processing")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Advantages: Zero external dependencies, complete customization,
batch processing capability")]
      }),

      new Paragraph({
        spacing: { before: 100, after: 50, left: 300 },
        children: [
          new TextRun({
            text: "INTERACTIVE HTML DASHBOARDS",
```

```
          bold: true,
          underline: {}
        })
      ]
    }),
    new Paragraph({
      spacing: { before: 50, after: 50, left: 600 },
      bullet: { level: 1 },
      children: [new TextRun("Use Case: Real-time data exploration and scenario modeling")]
    }),
    new Paragraph({
      spacing: { before: 50, after: 50, left: 600 },
      bullet: { level: 1 },
      children: [new TextRun("Technology Stack: React + D3.js + Chart.js for dynamic
visualizations")]
    }),
    new Paragraph({
      spacing: { before: 50, after: 50, left: 600 },
      bullet: { level: 1 },
      children: [new TextRun("Features: Drill-down analytics, what-if analysis, exportable
insights")
      ]
    }),

    new PageBreak(),

    // ============================================================
    // COMPLETE ATLAS PDD METHODOLOGY
    // ============================================================
    new Paragraph({
      heading: HeadingLevel.HEADING_1,
      spacing: { before: 400, after: 200 },
      children: [
        new TextRun({
          text: "COMPLETE ATLAS PROMPTWARE DESIGN METHODOLOGY",
          bold: true,
          size: 28,
          color: "2E75B6"
        })
      ]
    }),

    new Paragraph({
      heading: HeadingLevel.HEADING_2,
      spacing: { before: 300, after: 150 },
      children: [
        new TextRun({
          text: "3. ATLAS 5-Phase Decomposition Framework",
          bold: true,
          size: 24
```

```
          })
        ]
      }),

      new Paragraph({
        spacing: { before: 100, after: 100 },
        alignment: AlignmentType.JUSTIFIED,
        children: [
          new TextRun({
            text: "The ARK SI platform was systematically decomposed from a monolithic career
assessment concept into 98 discrete, testable, optimizable prompt modules following the ATLAS
(Adaptive Transformation & Logical Architecture System) methodology. This process ensures
repeatability, quality assurance, and operational excellence.",
            size: 22
          })
        ]
      }),

      // PHASE 1
      new Paragraph({
        heading: HeadingLevel.HEADING_3,
        spacing: { before: 200, after: 100 },
        children: [
          new TextRun({
            text: "Phase 1: Discovery & Requirements Analysis (Prompts 1-18)",
            bold: true,
            size: 22
          })
        ]
      }),

      new Paragraph({
        spacing: { before: 50, after: 50, left: 300 },
        bullet: { level: 0 },
        children: [
          new TextRun({
            text: "Core Modules:",
            bold: true
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-001: Resume/CV data extraction engine")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-002: LinkedIn profile intelligence mining")]
```

```
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-003: Manual assessment form processing")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-004: Career history normalization & standardization")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-005: Skills taxonomy mapping (O*NET integration)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-006: Industry classification (NAICS alignment)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-007-018: Data validation, completeness checks, edge case
handling")]
      }),

      // PHASE 2
      new Paragraph({
        heading: HeadingLevel.HEADING_3,
        spacing: { before: 200, after: 100 },
        children: [
          new TextRun({
            text: "Phase 2: JST Index Calculation Engine (Prompts 19-42)",
            bold: true,
            size: 22
          })
        ]
      }),

      new Paragraph({
        spacing: { before: 50, after: 50, left: 300 },
        bullet: { level: 0 },
        children: [
          new TextRun({
            text: "Jobs Layer Analysis:",
            bold: true
          })
        ]
```

```
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-019: Labor market demand scoring (job posting volume
analysis)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-020: Industry trajectory forecasting (growth/decline
patterns)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-021: Geographic market analysis (location-specific
demand)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-022: Experience curve positioning (years vs. market
average)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-023: Role saturation assessment (supply vs. demand
ratio)")]
      }),

      new Paragraph({
        spacing: { before: 100, after: 50, left: 300 },
        children: [
          new TextRun({
            text: "Skills Layer Analysis:",
            bold: true
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-024: Technical skills depth evaluation")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-025: Soft skills assessment (leadership, communication,
```

```
etc.)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-026: AI/Digital literacy scoring with future-readiness
bonus")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-027: Certification validation (industry-recognized
credentials)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-028: Cross-functional capability analysis")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-029: Skill currency assessment (recency & relevance)")]
      }),

      new Paragraph({
        spacing: { before: 100, after: 50, left: 300 },
        children: [
          new TextRun({
            text: "Talent Layer Analysis:",
            bold: true
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-030: Educational foundation scoring (degree level,
institution ranking)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-031: Professional achievement quality analysis")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-032: Career progression pattern evaluation")]
      }),
```

```
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-033: Learning agility indicators")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-034: Growth trajectory modeling")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-035-42: JST composite scoring, percentile ranking,
narrative generation")]
      }),

      // PHASE 3
      new Paragraph({
        heading: HeadingLevel.HEADING_3,
        spacing: { before: 200, after: 100 },
        children: [
          new TextRun({
            text: "Phase 3: AI Vulnerability Assessment (Prompts 43-64)",
            bold: true,
            size: 22
          })
        ]
      }),

      new Paragraph({
        spacing: { before: 50, after: 50, left: 300 },
        bullet: { level: 0 },
        children: [
          new TextRun({
            text: "5-Level Vulnerability Framework:",
            bold: true
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-043: Task-level automation potential analysis")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-044: AI capability maturity assessment (current AI tools
vs. job requirements)")]
```

```
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-045: Economic substitution viability (cost-benefit of
automation)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-046: Environmental barriers scoring (physical presence,
proprietary access, etc.)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-047: Timeline projection modeling (when will automation
be economically viable?)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-048-54: Level classification logic, risk mitigation
identification, heat mapping")]
      }),

      new Paragraph({
        spacing: { before: 100, after: 50, left: 300 },
        children: [
          new TextRun({
            text: "Protection Factor Analysis:",
            bold: true
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-055: Creative requirement quantification")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-056: Strategic decision-making complexity")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-057: Human relationship necessity scoring")]
      }),
```

```
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-058: Ethical judgment requirements")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-059-64: Composite vulnerability score, level assignment, 
narrative explanation")]
      }),

      // PHASE 4
      new Paragraph({
        heading: HeadingLevel.HEADING_3,
        spacing: { before: 200, after: 100 },
        children: [
          new TextRun({
            text: "Phase 4: Transferability Vector Analysis (Prompts 65-82)",
            bold: true,
            size: 22
          })
        ]
      }),

      new Paragraph({
        spacing: { before: 50, after: 50, left: 300 },
        bullet: { level: 0 },
        children: [
          new TextRun({
            text: "Adjacent Role Identification:",
            bold: true
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-065: O*NET similarity scoring (occupational 
relatedness)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-066: Skill overlap analysis (transferable vs. gap 
skills)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
```

```
        children: [new TextRun("PROMPT-067: Compensation premium calculation")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-068: Market demand comparison (current vs. target 
role)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-069: Transition timeline estimation")]
      }),

      new Paragraph({
        spacing: { before: 100, after: 50, left: 300 },
        children: [
          new TextRun({
            text: "Skill Gap Mapping:",
            bold: true
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-070: Required vs. current skill differential")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-071: Training pathway identification (courses, 
certifications, experience)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-072: Cost-to-close-gap estimation (tuition, time, 
opportunity cost)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-073: ROI modeling for each pivot opportunity")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-074-82: Network diagram generation, top-5 ranking, 
strategic recommendation synthesis")]
```

```
      }),

      // PHASE 5
      new Paragraph({
        heading: HeadingLevel.HEADING_3,
        spacing: { before: 200, after: 100 },
        children: [
          new TextRun({
            text: "Phase 5: Development Roadmap & Recommendations (Prompts 83-98)",
            bold: true,
            size: 22
          })
        ]
      }),

      new Paragraph({
        spacing: { before: 50, after: 50, left: 300 },
        bullet: { level: 0 },
        children: [
          new TextRun({
            text: "Upskilling Strategy Generation:",
            bold: true
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-083: New Skilling recommendations (entirely new
capabilities)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-084: Upskilling recommendations (enhancing existing
skills)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-085: Ready Skilling recommendations (AI tool adoption for
current role)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-086: Context Craft certification pathway mapping")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
```

```
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-087: Priority sequencing (high-ROI skills first)")]
      }),

      new Paragraph({
        spacing: { before: 100, after: 50, left: 300 },
        children: [
          new TextRun({
            text: "Actionable Timeline Creation:",
            bold: true
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-088: 30-60-90 day action plan generation")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-089: Milestone definition with success criteria")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-090: Checkpoint scheduling for progress tracking")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-091: Resource allocation guidance (time, budget, 
effort)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PROMPT-092-98: Final report compilation, visualization data 
export, social sharing optimization")]
      }),

      new PageBreak(),

      // ============================================================
      // ZPOS OPTIMIZATION METHODOLOGY
      // ============================================================
      new Paragraph({
        heading: HeadingLevel.HEADING_2,
        spacing: { before: 300, after: 150 },
        children: [
```

```
          new TextRun({
            text: "4. ZPOS Optimization Methodology",
            bold: true,
            size: 24
          })
        ]
      }),

      new Paragraph({
        spacing: { before: 100, after: 100 },
        alignment: AlignmentType.JUSTIFIED,
        children: [
          new TextRun({
            text: "The ARK SI system underwent comprehensive token optimization using ZPOS Level
3 methodology, achieving 45% token reduction while maintaining 97% semantic accuracy. This
optimization translates to 40-60% cost savings at enterprise scale while improving response
latency and throughput.",
            size: 22
          })
        ]
      }),

      new Paragraph({
        heading: HeadingLevel.HEADING_3,
        spacing: { before: 200, after: 100 },
        children: [
          new TextRun({
            text: "4.1 ZPOS Level 3 Compression Techniques Applied",
            bold: true,
            size: 22
          })
        ]
      }),

      new Paragraph({
        spacing: { before: 50, after: 50, left: 300 },
        bullet: { level: 0 },
        children: [
          new TextRun({
            text: "Lexical Compression:",
            bold: true
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Abbreviation standardization (e.g., 'years of experience' →
'YoE')")]
      }),
```

```
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Industry acronyms (NAICS, O*NET, DISC, JST)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Redundancy elimination (repeated instructional phrasing)")]
      }),

      new Paragraph({
        spacing: { before: 100, after: 50, left: 300 },
        children: [
          new TextRun({
            text: "Structural Compression:",
            bold: true
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("JSON schema replacement for verbose explanations")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Tabular data formatting instead of prose descriptions")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Conditional branching logic (if-then structures)")]
      }),

      new Paragraph({
        spacing: { before: 100, after: 50, left: 300 },
        children: [
          new TextRun({
            text: "Semantic Compression:",
            bold: true
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Context Craft 7-pillar framework reference (avoiding full pillar definitions in each prompt")]
```

```
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Shared global definitions (established once, referenced
throughout)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Implicit instruction (relying on model training for common
patterns)")]
      }),

      new Paragraph({
        heading: HeadingLevel.HEADING_3",
        spacing: { before: 200, after: 100 },
        children: [
          new TextRun({
            text: "4.2 Token Economics & Cost Savings Analysis",
            bold: true,
            size: 22
          })
        ]
      }),

      // Cost Table
      new Table({
        width: { size: 100, type: WidthType.PERCENTAGE },
        rows: [
          new TableRow({
            children: [
              new TableCell({ children: [new Paragraph({ text: "Metric", bold: true })] }),
              new TableCell({ children: [new Paragraph({ text: "Pre-Optimization", bold: true
})] }),
              new TableCell({ children: [new Paragraph({ text: "Post-ZPOS Level 3", bold: true
})] }),
              new TableCell({ children: [new Paragraph({ text: "Improvement", bold: true })] }),
            ],
          }),
          new TableRow({
            children: [
              new TableCell({ children: [new Paragraph("Total Prompt Tokens")] }),
              new TableCell({ children: [new Paragraph("23,520")] }),
              new TableCell({ children: [new Paragraph("12,936")] }),
              new TableCell({ children: [new Paragraph({ text: "45% reduction", color: "00B050"
})] }),
            ],
          }),
          new TableRow({
```

```
            children: [
              new TableCell({ children: [new Paragraph("Avg Response Time")] }),
              new TableCell({ children: [new Paragraph("4.8 seconds")] }),
              new TableCell({ children: [new Paragraph("2.6 seconds")] }),
              new TableCell({ children: [new Paragraph({ text: "46% faster", color: "00B050" })]
}),
            ],
          }),
          new TableRow({
            children: [
              new TableCell({ children: [new Paragraph("Cost per Assessment")] }),
              new TableCell({ children: [new Paragraph("$0.235")] }),
              new TableCell({ children: [new Paragraph("$0.129")] }),
              new TableCell({ children: [new Paragraph({ text: "45% lower cost", color: "00B050"
})] }),
            ],
          }),
          new TableRow({
            children: [
              new TableCell({ children: [new Paragraph("Annual Cost (10K assessments)")] }),
              new TableCell({ children: [new Paragraph("$2,350")] }),
              new TableCell({ children: [new Paragraph("$1,290")] }),
              new TableCell({ children: [new Paragraph({ text: "$1,060 saved", color: "00B050"
})] }),
            ],
          }),
          new TableRow({
            children: [
              new TableCell({ children: [new Paragraph("Semantic Accuracy")] }),
              new TableCell({ children: [new Paragraph("N/A (baseline)")] }),
              new TableCell({ children: [new Paragraph("97.0%")] }),
              new TableCell({ children: [new Paragraph({ text: "Quality maintained", color:
"00B050" })] }),
            ],
          }),
        ],
      }),

      new PageBreak(),

      // ============================================================
      // JCSE QUALITY METRICS
      // ============================================================
      new Paragraph({
        heading: HeadingLevel.HEADING_2,
        spacing: { before: 300, after: 150 },
        children: [
          new TextRun({
            text: "5. JCSE Quality Assurance Framework",
            bold: true,
```

```
            size: 24
          })
        ]
      }),

      new Paragraph({
        spacing: { before: 100, after: 100 },
        alignment: AlignmentType.JUSTIFIED,
        children: [
          new TextRun({
            text: "ARK SI achieved a Junglenomics Synergy Card Evaluation (JCSE) score of 48/50,
placing it in the 'FORGE Ultra Premium' category. This objective quality score across 10
dimensions demonstrates production-readiness and enterprise-grade reliability.",
            size: 22
          })
        ]
      }),

      new Paragraph({
        heading: HeadingLevel.HEADING_3,
        spacing: { before: 200, after: 100 },
        children: [
          new TextRun({
            text: "5.1 JCSE 10-Dimension Scoring Breakdown",
            bold: true,
            size: 22
          })
        ]
      }),

      // JCSE Scoring Table
      new Table({
        width: { size: 100, type: WidthType.PERCENTAGE },
        rows: [
          new TableRow({
            children: [
              new TableCell({ children: [new Paragraph({ text: "Dimension", bold: true })] }),
              new TableCell({ children: [new Paragraph({ text: "Score (0-5)", bold: true })] }),
              new TableCell({ children: [new Paragraph({ text: "Evaluation Criteria", bold: true
})] }),
            ],
          }),
          new TableRow({
            children: [
              new TableCell({ children: [new Paragraph("1. Clarity & Specificity")] }),
              new TableCell({ children: [new Paragraph("5/5")] }),
              new TableCell({ children: [new Paragraph("Unambiguous instructions, precise
terminology, zero interpretation needed")] }),
            ],
          }),
```

```
          new TableRow({
            children: [
              new TableCell({ children: [new Paragraph("2. Context Craft Integration")] }),
              new TableCell({ children: [new Paragraph("5/5")] }),
              new TableCell({ children: [new Paragraph("Full 7-pillar implementation (System,
Role, Instruction, Example, Constraint, Format, Data)")] }),
            ],
          }),
          new TableRow({
            children: [
              new TableCell({ children: [new Paragraph("3. Modularity & Composability")] }),
              new TableCell({ children: [new Paragraph("5/5")] }),
              new TableCell({ children: [new Paragraph("98 independent modules, clean
interfaces, zero dependencies between prompts")] }),
            ],
          }),
          new TableRow({
            children: [
              new TableCell({ children: [new Paragraph("4. Error Handling & Robustness")] }),
              new TableCell({ children: [new Paragraph("5/5")] }),
              new TableCell({ children: [new Paragraph("Comprehensive edge case coverage,
graceful degradation, validation at every stage")] }),
            ],
          }),
          new TableRow({
            children: [
              new TableCell({ children: [new Paragraph("5. Output Consistency")] }),
              new TableCell({ children: [new Paragraph("5/5")] }),
              new TableCell({ children: [new Paragraph("97% agreement across 5 trials,
deterministic scoring logic")] }),
            ],
          }),
          new TableRow({
            children: [
              new TableCell({ children: [new Paragraph("6. Token Efficiency")] }),
              new TableCell({ children: [new Paragraph("4/5")] }),
              new TableCell({ children: [new Paragraph("45% reduction achieved, ZPOS Level 3
applied, minor further optimization possible")] }),
            ],
          }),
          new TableRow({
            children: [
              new TableCell({ children: [new Paragraph("7. Real-World Applicability")] }),
              new TableCell({ children: [new Paragraph("5/5")] }),
              new TableCell({ children: [new Paragraph("Production-deployed at Fortune 500
scale, 8,742 assessments completed")] }),
            ],
          }),
          new TableRow({
            children: [
```

```
              new TableCell({ children: [new Paragraph("8. Documentation Quality")] }),
              new TableCell({ children: [new Paragraph("5/5")] }),
              new TableCell({ children: [new Paragraph("Complete ATLAS worksheet, PDD with usage
examples, API documentation")] }),
            ],
          }),
          new TableRow({
            children: [
              new TableCell({ children: [new Paragraph("9. Measurable Outcomes")] }),
              new TableCell({ children: [new Paragraph("5/5")] }),
              new TableCell({ children: [new Paragraph("Quantified ROI (8,955%), cost savings
($6M/year), satisfaction (4.6/5.0)")] }),
            ],
          }),
          new TableRow({
            children: [
              new TableCell({ children: [new Paragraph("10. Governance & Evolution")] }),
              new TableCell({ children: [new Paragraph("4/5")] }),
              new TableCell({ children: [new Paragraph("Lifecycle maturity protocols defined,
versioning implemented, autonomous operation pathway in progress")] }),
            ],
          }),
          new TableRow({
            children: [
              new TableCell({ children: [new Paragraph({ text: "TOTAL JCSE SCORE", bold: true
})] }),
              new TableCell({ children: [new Paragraph({ text: "48/50", bold: true, color:
"00B050" })] }),
              new TableCell({ children: [new Paragraph({ text: "FORGE Ultra Premium
Certification", bold: true })] }),
            ],
          }),
        ],
      }),

      new PageBreak(),

      // ============================================================
      // PRODUCTION DEPLOYMENT & RESULTS
      // ============================================================
      new Paragraph({
        heading: HeadingLevel.HEADING_1,
        spacing: { before: 400, after: 200 },
        children: [
          new TextRun({
            text: "PRODUCTION DEPLOYMENT & VALIDATED RESULTS",
            bold: true,
            size: 28,
            color: "2E75B6"
          })
```

```
  ]
}),

new Paragraph({
  heading: HeadingLevel.HEADING_2,
  spacing: { before: 300, after: 150 },
  children: [
    new TextRun({
      text: "6. Enterprise Deployment Case Study",
      bold: true,
      size: 24
    })
  ]
}),

new Paragraph({
  spacing: { before: 100, after: 50, left: 300 },
  children: [
    new TextRun({
      text: "Client Profile:",
      bold: true,
      underline: {}
    })
  ]
}),
new Paragraph({
  spacing: { before: 50, after: 50, left: 600 },
  bullet: { level: 1 },
  children: [new TextRun("Professional services consulting firm")]
}),
new Paragraph({
  spacing: { before: 50, after: 50, left: 600 },
  bullet: { level: 1 },
  children: [new TextRun("12,000 employees globally (North America, Europe, APAC)")]
}),
new Paragraph({
  spacing: { before: 50, after: 50, left: 600 },
  bullet: { level: 1 },
  children: [new TextRun("18% annual attrition (industry average: 15%)")]
}),
new Paragraph({
  spacing: { before: 50, after: 50, left: 600 },
  bullet: { level: 1 },
  children: [new TextRun("$350K average replacement cost per mid-level consultant")]
}),

new Paragraph({
  spacing: { before: 100, after: 50, left: 300 },
  children: [
    new TextRun({
```

```
            text: "Challenge:",
            bold: true,
            underline: {}
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Employees uncertain about career value in AI-augmented
economy")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("HR unable to quantify skill gaps systematically")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Career development planning ad-hoc and inconsistent")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Talent retention declining (67% of exiters cited 'lack of career
growth visibility')")]
      }),

      new Paragraph({
        heading: HeadingLevel.HEADING_3,
        spacing: { before: 200, after: 100 },
        children: [
          new TextRun({
            text: "6.1 Quantified Business Impact (12-Month Post-Deployment)",
            bold: true,
            size: 22
          })
        ]
      }),

      // Results Table
      new Table({
        width: { size: 100, type: WidthType.PERCENTAGE },
        rows: [
          new TableRow({
            children: [
              new TableCell({ children: [new Paragraph({ text: "Metric", bold: true })] }),
              new TableCell({ children: [new Paragraph({ text: "Pre-ARK SI", bold: true })] }),
              new TableCell({ children: [new Paragraph({ text: "Post-ARK SI", bold: true })] }),
```

```
            new TableCell({ children: [new Paragraph({ text: "Impact", bold: true })] }),
          ],
        }),
        new TableRow({
          children: [
            new TableCell({ children: [new Paragraph("Attrition Rate")] }),
            new TableCell({ children: [new Paragraph("18%")] }),
            new TableCell({ children: [new Paragraph("11%")] }),
            new TableCell({ children: [new Paragraph({ text: "39% improvement", color:
"00B050" })] }),
          ],
        }),
        new TableRow({
          children: [
            new TableCell({ children: [new Paragraph("Career Coaching Time")] }),
            new TableCell({ children: [new Paragraph("4-6 hours/person")] }),
            new TableCell({ children: [new Paragraph("12 minutes/person")] }),
            new TableCell({ children: [new Paragraph({ text: "96% faster", color: "00B050" })]
}),
          ],
        }),
        new TableRow({
          children: [
            new TableCell({ children: [new Paragraph("Assessments Completed")] }),
            new TableCell({ children: [new Paragraph("N/A")] }),
            new TableCell({ children: [new Paragraph("8,742")] }),
            new TableCell({ children: [new Paragraph("73% workforce coverage")] }),
          ],
        }),
        new TableRow({
          children: [
            new TableCell({ children: [new Paragraph("Repeat Usage Rate")] }),
            new TableCell({ children: [new Paragraph("N/A")] }),
            new TableCell({ children: [new Paragraph("64% quarterly")] }),
            new TableCell({ children: [new Paragraph("High engagement")] }),
          ],
        }),
        new TableRow({
          children: [
            new TableCell({ children: [new Paragraph("User Satisfaction")] }),
            new TableCell({ children: [new Paragraph("N/A")] }),
            new TableCell({ children: [new Paragraph("4.6/5.0")] }),
            new TableCell({ children: [new Paragraph("92% approval")] }),
          ],
        }),
        new TableRow({
          children: [
            new TableCell({ children: [new Paragraph("Training Course Completion")] }),
            new TableCell({ children: [new Paragraph("47%")] }),
            new TableCell({ children: [new Paragraph("63%")] }),
```

```
            new TableCell({ children: [new Paragraph({ text: "34% increase", color: "00B050"
})] }),
          ],
        }),
        new TableRow({
          children: [
            new TableCell({ children: [new Paragraph("Internal Mobility (lateral moves)")] }),
            new TableCell({ children: [new Paragraph("82 moves/year")] }),
            new TableCell({ children: [new Paragraph("105 moves/year")] }),
            new TableCell({ children: [new Paragraph({ text: "28% increase", color: "00B050"
})] }),
          ],
        }),
        new TableRow({
          children: [
            new TableCell({ children: [new Paragraph("Skill Gap Closure Time")] }),
            new TableCell({ children: [new Paragraph("8.2 months avg")] }),
            new TableCell({ children: [new Paragraph("4.8 months avg")] }),
            new TableCell({ children: [new Paragraph({ text: "41% faster", color: "00B050" })]
}),
          ],
        }),
      ],
    }),

    new Paragraph({
      heading: HeadingLevel.HEADING_3,
      spacing: { before: 200, after: 100 },
      children: [
        new TextRun({
          text: "6.2 Economic Impact Analysis",
          bold: true,
          size: 22
        })
      ]
    }),

    new Paragraph({
      spacing: { before: 50, after: 50, left: 300 },
      bullet: { level: 0 },
      children: [
        new TextRun({
          text: "Attrition Cost Savings:",
          bold: true
        })
      ]
    }),
    new Paragraph({
      spacing: { before: 50, after: 50, left: 600 },
      bullet: { level: 1 },
```

```
          children: [new TextRun("Baseline: 18% × 12,000 = 2,160 departures/year")]
        }),
        new Paragraph({
          spacing: { before: 50, after: 50, left: 600 },
          bullet: { level: 1 },
          children: [new TextRun("Post-ARK SI: 11% × 12,000 = 1,320 departures/year")]
        }),
        new Paragraph({
          spacing: { before: 50, after: 50, left: 600 },
          bullet: { level: 1 },
          children: [new TextRun("Reduction: 840 departures prevented")]
        }),
        new Paragraph({
          spacing: { before: 50, after: 50, left: 600 },
          bullet: { level: 1 },
          children: [new TextRun({ text: "Annual Savings: 840 × $350K = $4.2M", bold: true, color: "00B050" })]
        }),

        new Paragraph({
          spacing: { before: 100, after: 50, left: 300 },
          children: [
            new TextRun({
              text: "Training ROI Improvement:",
              bold: true
            })
          ]
        }),
        new Paragraph({
          spacing: { before: 50, after: 50, left: 600 },
          bullet: { level: 1 },
          children: [new TextRun("Completion rate increase: 47% → 63% (34% improvement)")]
        }),
        new Paragraph({
          spacing: { before: 50, after: 50, left: 600 },
          bullet: { level: 1 },
          children: [new TextRun("Additional completions: 1,920 courses/year")]
        }),
        new Paragraph({
          spacing: { before: 50, after: 50, left: 600 },
          bullet: { level: 1 },
          children: [new TextRun("Avg value per completed course: $950")]
        }),
        new Paragraph({
          spacing: { before: 50, after: 50, left: 600 },
          bullet: { level: 1 },
          children: [new TextRun({ text: "Additional Value Realized: $1.8M", bold: true, color: "00B050" })]
        }),
```

```
new Paragraph({
  spacing: { before: 100, after: 50, left: 300 },
  children: [
    new TextRun({
      text: "Implementation Cost:",
      bold: true
    })
  ]
}),
new Paragraph({
  spacing: { before: 50, after: 50, left: 600 },
  bullet: { level: 1 },
  children: [new TextRun("ATLAS PDD development: $45K (10 weeks)")]
}),
new Paragraph({
  spacing: { before: 50, after: 50, left: 600 },
  bullet: { level: 1 },
  children: [new TextRun("Platform integration: $12K")]
}),
new Paragraph({
  spacing: { before: 50, after: 50, left: 600 },
  bullet: { level: 1 },
  children: [new TextRun("Training & rollout: $10K")]
}),
new Paragraph({
  spacing: { before: 50, after: 50, left: 600 },
  bullet: { level: 1 },
  children: [new TextRun({ text: "Total Investment: $67K", bold: true })]
}),

new Paragraph({
  spacing: { before: 100, after: 100, left: 300 },
  shading: { fill: "E7F4E4" },
  border: {
    top: { style: BorderStyle.DOUBLE, size: 3, color: "00B050" },
    bottom: { style: BorderStyle.DOUBLE, size: 3, color: "00B050" },
    left: { style: BorderStyle.DOUBLE, size: 3, color: "00B050" },
    right: { style: BorderStyle.DOUBLE, size: 3, color: "00B050" }
  },
  children: [
    new TextRun({
      text: "TOTAL 1-YEAR VALUE: $6.0M (attrition + training)",
      bold: true,
      size: 24,
      color: "00B050"
    })
  ]
}),

new Paragraph({
```

```
  spacing: { before: 50, after: 100, left: 300 },
  shading: { fill: "E7F4E4" },
  border: {
    top: { style: BorderStyle.DOUBLE, size: 3, color: "00B050" },
    bottom: { style: BorderStyle.DOUBLE, size: 3, color: "00B050" },
    left: { style: BorderStyle.DOUBLE, size: 3, color: "00B050" },
    right: { style: BorderStyle.DOUBLE, size: 3, color: "00B050" }
  },
  children: [
    new TextRun({
      text: "ROI: 8,955% ($6M value ÷ $67K investment)",
      bold: true,
      size: 24,
      color: "00B050"
    })
  ]
}),

new PageBreak(),

// ============================================================
// VISUALIZATION ENGINE TECHNICAL SPECIFICATIONS
// ============================================================
new Paragraph({
  heading: HeadingLevel.HEADING_1,
  spacing: { before: 400, after: 200 },
  children: [
    new TextRun({
      text: "VISUALIZATION ENGINE TECHNICAL SPECIFICATIONS",
      bold: true,
      size: 28,
      color: "2E75B6"
    })
  ]
}),

new Paragraph({
  heading: HeadingLevel.HEADING_2,
  spacing: { before: 300, after: 150 },
  children: [
    new TextRun({
      text: "7. Multi-Platform Rendering Architecture",
      bold: true,
      size: 24
    })
  ]
}),

new Paragraph({
  spacing: { before: 100, after: 100 },
```

```
        alignment: AlignmentType.JUSTIFIED,
        children: [
          new TextRun({
            text: "ARK SI employs a multi-tiered visualization strategy leveraging best-in-class tools for different use cases, ensuring optimal quality, performance, and cost-efficiency across deployment scenarios.",
            size: 22
          })
        ]
      }),

      new Paragraph({
        heading: HeadingLevel.HEADING_3,
        spacing: { before: 200, after: 100 },
        children: [
          new TextRun({
            text: "7.1 NotebookLM Integration (Primary Automated Generator)",
            bold: true,
            size: 22
          })
        ]
      }),

      new Paragraph({
        spacing: { before: 50, after: 50, left: 300 },
        bullet: { level: 0 },
        children: [new TextRun({ text: "Technology: Google Nano Banana Pro model (Gemini-based)", bold: true })]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 300 },
        bullet: { level: 0 },
        children: [new TextRun({ text: "Deployment: November 2025 production release", bold: true })]
      }),

      new Paragraph({
        spacing: { before: 100, after: 50, left: 600 },
        children: [
          new TextRun({
            text: "Capabilities:",
            bold: true,
            underline: {}
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 900 },
        bullet: { level: 2 },
        children: [new TextRun("Automated infographic generation from structured markdown
```

```
sources")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 900 },
        bullet: { level: 2 },
        children: [new TextRun("Three orientation options: Portrait (9:16), Landscape (16:9),
Square (1:1)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 900 },
        bullet: { level: 2 },
        children: [new TextRun("Detail levels: Concise, Standard, Detailed (beta)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 900 },
        bullet: { level: 2 },
        children: [new TextRun("Custom prompt engineering for brand alignment")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 900 },
        bullet: { level: 2 },
        children: [new TextRun("30-second generation time (avg)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 900 },
        bullet: { level: 2 },
        children: [new TextRun("PNG export at 300 DPI (print-quality)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 900 },
        bullet: { level: 2 },
        children: [new TextRun("Public sharing links for social distribution")]
      }),

      new Paragraph({
        spacing: { before: 100, after: 50, left: 600 },
        children: [
          new TextRun({
            text: "Integration Workflow:",
            bold: true,
            underline: {}
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 900 },
        bullet: { level: 2 },
        children: [new TextRun("1. ARK SI generates structured JSON/markdown assessment
output")]
      }),
```

```
      new Paragraph({
        spacing: { before: 50, after: 50, left: 900 },
        bullet: { level: 2 },
        children: [new TextRun("2. Data formatter converts to NotebookLM-compatible format")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 900 },
        bullet: { level: 2 },
        children: [new TextRun("3. Custom prompt template applies Junglenomics branding
specifications")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 900 },
        bullet: { level: 2 },
        children: [new TextRun("4. Nano Banana Pro generates infographic (30-90 seconds)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 900 },
        bullet: { level: 2 },
        children: [new TextRun("5. PNG downloaded via NotebookLM interface")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 900 },
        bullet: { level: 2 },
        children: [new TextRun("6. Automated embedding into comprehensive PDF report")]
      }),

      new Paragraph({
        heading: HeadingLevel.HEADING_3,
        spacing: { before: 200, after: 100 },
        children: [
          new TextRun({
            text: "7.2 Native PDF Generation (Python ReportLab)",
            bold: true,
            size: 22
          })
        ]
      }),

      new Paragraph({
        spacing: { before: 50, after: 50, left: 300 },
        bullet: { level: 0 },
        children: [new TextRun({ text: "Technology Stack:", bold: true })]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("ReportLab 4.0+ (core PDF generation)")]
      }),
      new Paragraph({
```

```
          spacing: { before: 50, after: 50, left: 600 },
          bullet: { level: 1 },
          children: [new TextRun("Matplotlib 3.8+ (charts, graphs, radar diagrams)")]
        }),
        new Paragraph({
          spacing: { before: 50, after: 50, left: 600 },
          bullet: { level: 1 },
          children: [new TextRun("NetworkX 3.2+ (transferability network visualizations)")]
        }),
        new Paragraph({
          spacing: { before: 50, after: 50, left: 600 },
          bullet: { level: 1 },
          children: [new TextRun("Pandas 2.1+ (data processing and tabular formatting)")]
        }),
        new Paragraph({
          spacing: { before: 50, after: 50, left: 600 },
          bullet: { level: 1 },
          children: [new TextRun("Pillow 10.1+ (image processing and optimization)")]
        }),

        new Paragraph({
          spacing: { before: 100, after: 50, left: 300 },
          children: [
            new TextRun({
              text: "Report Structure (Multi-Page Format):",
              bold: true
            })
          ]
        }),
        new Paragraph({
          spacing: { before: 50, after: 50, left: 600 },
          bullet: { level: 1 },
          children: [new TextRun("Page 1: Single-page visual summary (6-section layout per
specifications)")]
        }),
        new Paragraph({
          spacing: { before: 50, after: 50, left: 600 },
          bullet: { level: 1 },
          children: [new TextRun("Page 2-3: Detailed JST Index analysis with layer breakdowns")]
        }),
        new Paragraph({
          spacing: { before: 50, after: 50, left: 600 },
          bullet: { level: 1 },
          children: [new TextRun("Page 4-5: AI Vulnerability deep-dive with task-level heat
mapping")]
        }),
        new Paragraph({
          spacing: { before: 50, after: 50, left: 600 },
          bullet: { level: 1 },
          children: [new TextRun("Page 6-7: Transferability vectors with skill gap matrices")]
```

```
    }),
    new Paragraph({
      spacing: { before: 50, after: 50, left: 600 },
      bullet: { level: 1 },
      children: [new TextRun("Page 8-10: Personalized roadmap with 30-60-90 day action plan")]
    }),
    new Paragraph({
      spacing: { before: 50, after: 50, left: 600 },
      bullet: { level: 1 },
      children: [new TextRun("Page 11: Context Craft certification pathway visualization")]
    }),
    new Paragraph({
      spacing: { before: 50, after: 50, left: 600 },
      bullet: { level: 1 },
      children: [new TextRun("Page 12: Resources, references, and next steps")]
    }),

    new Paragraph({
      heading: HeadingLevel.HEADING_3,
      spacing: { before: 200, after: 100 },
      children: [
        new TextRun({
          text: "7.3 Interactive Dashboard (React + D3.js)",
          bold: true,
          size: 22
        })
      ]
    }),

    new Paragraph({
      spacing: { before: 50, after: 50, left: 300 },
      bullet: { level: 0 },
      children: [new TextRun({ text: "Technology Stack:", bold: true })]
    }),
    new Paragraph({
      spacing: { before: 50, after: 50, left: 600 },
      bullet: { level: 1 },
      children: [new TextRun("React 18+ (component architecture)")]
    }),
    new Paragraph({
      spacing: { before: 50, after: 50, left: 600 },
      bullet: { level: 1 },
      children: [new TextRun("D3.js 7+ (dynamic data visualizations)")]
    }),
    new Paragraph({
      spacing: { before: 50, after: 50, left: 600 },
      bullet: { level: 1 },
      children: [new TextRun("Chart.js 4+ (interactive charts and gauges)")]
    }),
    new Paragraph({
```

```
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Recharts 2+ (responsive chart library)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Tailwind CSS (responsive styling framework)")]
      }),

      new Paragraph({
        spacing: { before: 100, after: 50, left: 300 },
        children: [
          new TextRun({
            text: "Interactive Features:",
            bold: true
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Drill-down analytics (click JST layers to see component
scores)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("What-if scenario modeling (adjust skills, see impact on
vulnerability)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Transferability network exploration (hover nodes for details)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Timeline roadmap customization (prioritize recommendations)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Export to PDF/PNG from any view")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Share via link (requires authentication)")]
```

```
}),

new PageBreak(),

// ============================================================
// IMPLEMENTATION ROADMAP
// ============================================================
new Paragraph({
  heading: HeadingLevel.HEADING_1,
  spacing: { before: 400, after: 200 },
  children: [
    new TextRun({
      text: "IMPLEMENTATION ROADMAP & NEXT STEPS",
      bold: true,
      size: 28,
      color: "2E75B6"
    })
  ]
}),

new Paragraph({
  heading: HeadingLevel.HEADING_2,
  spacing: { before: 300, after: 150 },
  children: [
    new TextRun({
      text: "8. Deployment Strategy",
      bold: true,
      size: 24
    })
  ]
}),

new Paragraph({
  heading: HeadingLevel.HEADING_3,
  spacing: { before: 200, after: 100 },
  children: [
    new TextRun({
      text: "8.1 Phased Rollout Timeline",
      bold: true,
      size: 22
    })
  ]
}),

new Paragraph({
  spacing: { before: 50, after: 50, left: 300 },
  children: [
    new TextRun({
      text: "PHASE 1: Foundation (Weeks 1-4)",
      bold: true,
```

```
            underline: {}
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Platform infrastructure setup (cloud hosting, databases,
APIs)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("NotebookLM integration testing and custom prompt refinement")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("PDF generation pipeline validation")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Beta testing with 50 internal users")]
      }),

      new Paragraph({
        spacing: { before: 100, after: 50, left: 300 },
        children: [
          new TextRun({
            text: "PHASE 2: Pilot Deployment (Weeks 5-12)",
            bold: true,
            underline: {}
          })
        ]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Rollout to 500 users (single business unit)")]
      }),
      new Paragraph({
        spacing: { before: 50, after: 50, left: 600 },
        bullet: { level: 1 },
        children: [new TextRun("Gather feedback, refine UX, optimize performance")]
      }),
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```

```
        }),
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      // ============================================================
      // CONCLUSION & APPENDICES
      // ============================================================
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prompt engineering when combined with state-of-the-art visualization technologies. By
integrating the ATLAS PromptWare methodology with NotebookLM's Nano Banana Pro model, native PDF
generation, and interactive dashboards, ARK SI delivers consulting-grade career intelligence at
unprecedented scale and efficiency. The platform's validated 8,955% ROI, 39% attrition
reduction, and 4.6/5.0 satisfaction score demonstrate that AI vulnerability is not a threat to
be feared but an opportunity to be strategically navigated. As the AI-augmented economy
continues to evolve, ARK SI positions professionals and organizations alike to thrive through
```

```
data-driven career resilience, systematic upskilling, and proactive talent development. The
integration of visualization platforms like NotebookLM represents a paradigm shift from manual
design work to automated intelligence generation, enabling career coaches, HR professionals, and
individuals to access sophisticated analytics previously reserved for Fortune 500 consultancies.
ARK SI is not merely a career assessment tool—it is the definitive platform for quantifying
human capital value in the age of AI.",
            size: 22
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        ]
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      // ================================================================
      // DOCUMENT METADATA & CERTIFICATION
      // ================================================================
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}),
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Integrated Visualization Edition")] }),
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```
            ],
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          new TableRow({
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Amusa (Founder & AI Product Design Architect)")] }),
            ],
          }),
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Craft Academy")] }),
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```

```
            new TableCell({ children: [new Paragraph("Idea Factory |
ideafactory@junglenomics.ai | www.junglenomics.ai")] }),
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      ],
    }),

    new Paragraph({
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    }),

    new Paragraph({
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          size: 16
        })
      ]
    }),
  ]
 }]
});

// Generate the document
Packer.toBuffer(doc).then(buffer => {
```

```
fs.writeFileSync("/home/claude/ARK_SI_PromptWare_Design_Document_v2.0_INTEGRATED_VISUALIZATION.d
ocx", buffer);
  console.log("✓ Document created successfully!");
  console.log("✓ Filename:
ARK_SI_PromptWare_Design_Document_v2.0_INTEGRATED_VISUALIZATION.docx");
});
```